

Adamyia Singh

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EDUCATION

Rutgers University

New Brunswick, NJ

B.S. in Computer Science and Data Science | GPA: 3.6

Aug 2023 – May 2027

- **Relevant Coursework:** Machine Learning, Algorithms, Data Structures, Computer Architecture
- **Leadership & Activities:** USACS (CS Club), STAR (Space Technology Association at Rutgers)
- **Honors & Awards:** Winner, Verizon Smart Campus Challenge 2025; Dean's List; Eagle Scout

EXPERIENCE

Machine Learning Engineer Intern

Nov. 2025 – Present

Aviro (YC-backed enterprise AI agents platform)

Remote, San Francisco, CA

- Built a Dockerized **Enterprise Search Benchmark** (10 tasks) over a corpus of PDFs/PowerPoints/spreadsheets, backed by **PostgreSQL + pgvector** semantic search
- Designed an **MCP-based RL environment** exposing **search/fetch/answer** tools with step budgeting, citation validation, and redaction-after-view memory constraints, improving model performance by **26%**
- Trained **Qwen3-14B** with **GRPO** via (1) W&B serverless distillation from frontier-model trajectories and (2) local on-policy RL using **vLLM + LoRA** with hot-reloaded adapters

Incoming Software Engineering Intern – Data Engineering

Jun. 2026 – Aug. 2026

Moody's Corporation

New York, NY

Software Engineer (MVP Development), Fusen Fellow

May 2025 – Aug. 2025

rebootED (tryrebooted.com, backed by Fusen World Accelerator & Google for Startups)

Atlanta, GA

- Engineered an enterprise L&D MVP end-to-end: designed **Spring Boot + PostgreSQL** services with JWT auth and REST APIs, and a **Next.js + TypeScript** frontend with role-based dashboards and analytics
- Implemented AI-driven course generation pipelines

Undergraduate Research Assistant - Computer Vision

Jan. 2025 – Present

Rutgers University, Computer Science Department

New Brunswick, NJ

- Benchmarked **3+ GenAI VLMs** (e.g., **CLIP**, **BioMedCLIP**) on negation-sensitive tasks using **PyTorch** and **225K+ multimodal medical images** from the CheXpert dataset

PROJECTS

SO-ARM-101 MuJoCo Simulation | *MuJoCo, Gymnasium, PyTorch, Transformers, RL*

- Built a MuJoCo + **Gymnasium** simulation environment for the SO-ARM-101 arm with multi-camera observations, normalized state/action interfaces, teleoperation, and shaped pick-and-place rewards
- Implemented **VLA policy inference** with adapters for **SmolVLA (450M)** and **Pi0 (3.3B)**, enabling model swapping via CLI flags and consistent action formatting
- Trained policies with **ReinFlow** using PPO-style objectives with exact log-probs; supported headless + parallel rollouts, checkpointing, and inference from trained policies

RU-Networking | *Next.js, OpenAI GPT-4o, Assistant API, RAG, PineconeDB*

- Built a full-stack AI-powered course planning assistant designed to support **70,000+** Rutgers students using RAG, engineering function-calling workflows, tool orchestration, and Server-Sent Events to deliver **real-time, production-scale chat recommendations**.
- Reverse-engineered undocumented endpoints and developed a highly parallelized Python scraper to extract and clean a **1M+ line dataset**, powering a custom SQL-backed recommendation engine.

TECHNICAL SKILLS

Languages: Python, C/C++, Java, TypeScript, React.js, SQL, Bash

ML: PyTorch, HuggingFace, vLLM, LoRA, GRPO, DSPy, Transformers, RAG, NLP, VLA, Computer Vision, Reinforcement Learning

Infra / Backend: FastAPI, Spring Boot, PostgreSQL, pgvector, Docker, Git, AWS S3, Linux, CUDA, REST APIs